

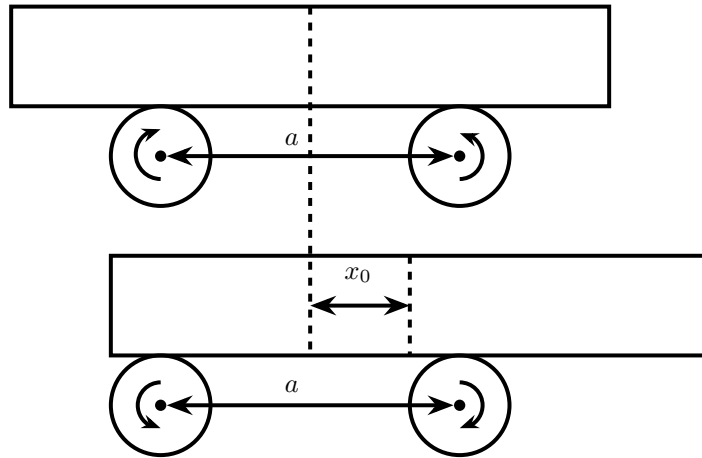
Physics Olympiad L2-22

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Problem 1: Rolling Plank

A uniform thin rigid rod of mass M is supported by two rotating rollers whose axes are separated by a fixed distance a . The rod is initially placed at rest asymmetrically at a distance x_0 away from the equilibrium position. The coefficient of friction between the bar and the rollers is μ . Determine the displacement of the centre of mass of the rod $x(t)$ when:

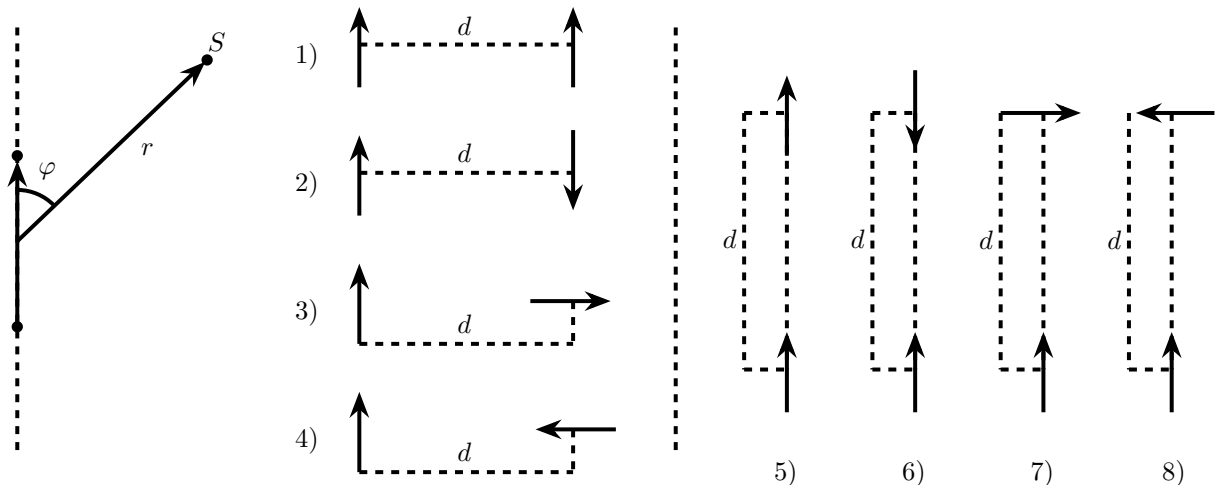
- The rollers rotate in opposite direction as shown in the figure;
- The rollers rotate in reverse directions to the ones shown in the figure;



Problem 2: Interacting Dipoles

Assume you are given a dipole with known charge q and length l , or equivalently the dipole moment $\vec{p}$.

- Find an expression for the potential φ of the dipole at point S as shown on the figure below;
- Using this expression, find the polar components of the electric field strength vector;
- Find the points at which $\vec{E}$ is perpendicular to $\vec{p}$;
- The dipole is placed in an external uniform electric field whose strength is $\vec{E}_0 \uparrow \uparrow \vec{p}$. In this case one of the equipotential surfaces enclosing the dipole forms a sphere. Determine the radius of this sphere;
- In figures 1) through 8) two identical dipoles are placed in different arrangements as shown, with their centres being a distance $d \gg l$ from one another in each case. For each arrangement shown find approximated expressions for the energy of interaction E_p and the force of interaction $\vec{F}$.



Problem 3: Wave Superposition

Two plane waves ψ_1 and ψ_2 propagate without energy loss in a homogeneous elastic medium, one along the x -axis and the other along the y -axis:

$$\rightarrow \psi_1(x, t) = a \cos(\omega t - kx);$$

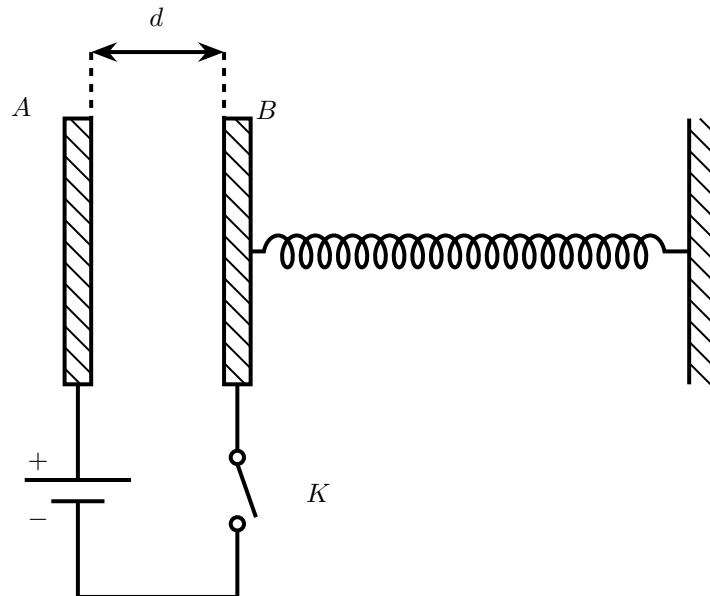
$$\rightarrow \psi_2(x, t) = a \cos(\omega t - ky);$$

Find the wave motion pattern of the particles in the plane xy if both waves are:

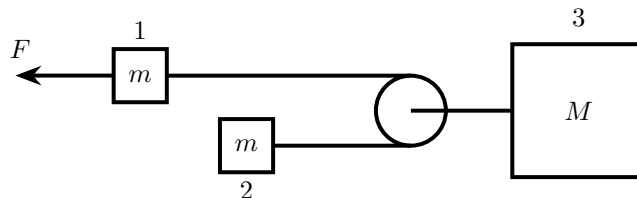
- transverse and their oscillation directions coincide;
- longitudinal.

Problem 4: Oscillating Capacitor

Plate A of a parallel-plate capacitor is fixed, while plate B is attached to a wall with a spring and can move while remaining parallel to plate A . After closing the switch K , plate B starts moving and eventually stops in a new equilibrium position. In this case, the initial equilibrium distance d between the capacitor plates (when the spring is not stretched) decreases by $\eta = 10\%$. How much would the equilibrium distance between the plates change if the switch K were closed for a short time? Assume that during this time, plate B does not have time to move significantly.

**Problem 5: Body-Pulley System**

Find the acceleration of the bodies in the system shown in the figure. The force F is applied along the direction of the string to one of the bodies of mass m . The sections of the string on both sides of the light pulley, which is attached to a body of mass M , are parallel.

**Problem 6: Battery Efficiency**

A battery consists of N identical cells, each of e.m.f. ε . Is it true that the energy wasted when using the battery to charge a capacitor through a resistor can be reduced by charging it in N stages? That is by connecting it first across a single cell, and then across two cells, and so on, rather than across the whole battery in a single step.

Appendix

Useful Mathematical Formulae and Physical Constants

Important: Infinite series that are marked with (*) are only valid under the condition that $|x| < 1$.

Taylor Expansions

$$\begin{aligned}\sin(x) &= x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(x^7) \\ \cos(x) &= 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \mathcal{O}(x^8) \\ \operatorname{tg}(x) &= x + \frac{x^3}{3} + \frac{2x^5}{15} + \mathcal{O}(x^7) \\ \arcsin(x) &= x + \frac{x^3}{6} + \frac{3x^5}{40} + \mathcal{O}(x^7) \\ \arctg(x) &= x - \frac{x^3}{3} + \frac{x^5}{5} + \mathcal{O}(x^7) \\ \exp(x) &= 1 + x + \frac{x^2}{2} + \mathcal{O}(x^3) \\ \ln(1+x) &= x - \frac{1}{2}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1-x)^{-1} &= 1 + x + x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{-1} &= 1 - x + x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{+\frac{1}{2}} &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{-\frac{1}{2}} &= 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{+\frac{3}{2}} &= 1 + \frac{3}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \quad (*) \\ (1+x)^{-\frac{3}{2}} &= 1 - \frac{3}{2}x + \frac{15}{8}x^2 + \mathcal{O}(x^3) \quad (*)\end{aligned}$$

Inverse Hyperbolic Functions

$$\begin{aligned}\operatorname{arsinh}(x) &= \ln(x + \sqrt{x^2 + 1}) \\ \operatorname{arcosh}(x) &= \ln(x + \sqrt{x^2 - 1}) \\ \operatorname{artgh}(x) &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \\ \operatorname{arctgh}(x) &= \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right)\end{aligned}$$

Indefinite Integrals

$$\begin{aligned}\int (1-x^2)^{-\frac{3}{2}} dx &= \frac{x}{\sqrt{1-x^2}} + C \\ \int (1+x^2)^{-\frac{3}{2}} dx &= \frac{x}{\sqrt{1+x^2}} + C \\ \int (1-x^2)^{-\frac{1}{2}} dx &= \arcsin(x) + C \\ \int (1+x^2)^{-\frac{1}{2}} dx &= \operatorname{arsinh}(x) + C \\ \int (1-x^2)^{+\frac{1}{2}} dx &= \frac{\arcsin(x) + x\sqrt{1-x^2}}{2} + C \\ \int (1+x^2)^{+\frac{1}{2}} dx &= \frac{\operatorname{arsinh}(x) + x\sqrt{1+x^2}}{2} + C\end{aligned}$$

Useful Physical Constants

Constant	Symbol	Value	Measuring Units
Light speed in vacuum	c	$3,000 \times 10^8$	m/s
Rest mass of electron	m_e	$9,109 \times 10^{-31}$	kg
Rest mass of proton	m_p	$1,673 \times 10^{-27}$	kg
Rest mass of neutron	m_n	$1,675 \times 10^{-27}$	kg
Rest mass of muon	m_μ	$1,884 \times 10^{-28}$	kg
Rest mass of tau-lepton	m_τ	$3,168 \times 10^{-27}$	kg
Elementary charge	e	$1,602 \times 10^{-19}$	C
Planck's constant	h	$6,626 \times 10^{-34}$	Js
Boltzmann's constant	k	$1,381 \times 10^{-23}$	J/K
Gravitational constant	γ	$6,673 \times 10^{-11}$	Nm ² /kg ²
Dielectric permittivity of vacuum	ε_0	$8,854 \times 10^{-12}$	F/m
Avogadro's number	N_A	$6,022 \times 10^{23}$	mol ⁻¹
Rydberg's constant	R_∞	$1,097 \times 10^7$	m ⁻¹
Magnetic permeability of vacuum	μ_0	$4\pi \times 10^{-7}$	H/m
Universal gas constant	R	8,314	J/(mol · K)